

On-Chip Sensing with Slow Light Topological Cavities

Bodhan Chakraborty, Wenhao Wang, Nikhil Navaratna, Yi Ji Tan, and Ranjan Singh*

Terahertz (THz) on-chip sensing has emerged as a powerful approach for high spatial resolution, real-time, label-free, and non-invasive detection. Valley photonic crystals (VPC)-based waveguide-cavity systems featuring topologically protected signal propagation with reduced scattering losses can pave the way for robust THz sensing platforms. However, achieving high detection sensitivity through strong light-matter interactions within a compact device footprint remains challenging. Slow-light effects enable spatial compression of optical energy and enhance light-matter interaction time by reducing the group velocity, thereby improving sensitivity within a reduced cavity dimension. In this work, we present an on-chip slow-light VPC waveguide-cavity system designed for high-detection sensitivity. We demonstrate that slow-light engineering effectively compensates for the reduced physical dimensions of the slow-light cavities by exploiting dispersion and modal control. This approach shortens the round-trip path length by up to 1.76 times compared with the non-slow-light cavities, thereby enhancing light-matter interactions within a smaller cavity footprint. Through polyimide thin-film sensing experiments, we demonstrate that slow-light VPC waveguide-cavity systems exhibit up to a threefold increase in sensitivity and threefold reduction in thickness resolution compared to systems without slow-light effects. Our work offers a novel platform for next-generation, compact, and high-performance THz sensing.

1. Introduction

The terahertz (THz) frequency range (0.1–10 THz) holds significant promise for on-chip sensing, owing to its millimeter- to sub-millimeter-scale wavelengths, which offer superior spatial resolution compared to microwaves and lower-millimeter wave bands.^[1] In addition, THz waves offer greater penetration depths than visible and near-infrared light, enabling the analysis of a broader range of analytes. Furthermore, their sensitivity to weak hydrogen bonds allows for the detection of distinct molecular spectral signatures,^[2,3] while their non-ionizing nature (4.14 meV photon energy at 1 THz) makes them ideal for safe, high-resolution sensing, particularly relevant for next-generation network integrated sensing applications in sixth-generation (6G) communication systems.^[4–7] Consequently, a wide range of functionalized THz on-chip sensing platforms have been demonstrated, including microstrip lines, photonic crystal-based resonator cavities, and microcapillary-integrated photonic crystal resonators.^[8–11] However, these conventional on-chip sensors face several challenges that limit their effectiveness

and scalability. Specifically, their performance is often compromised by material defects, fabrication imperfections, surface roughness, and external noise, all of which degrade sensitivity. In this context, topological photonic structures present a compelling alternative platform.^[12–14] Owing to inherent topological protection, these systems exhibit immunity to backscattering and structural imperfections, allowing light to propagate robustly along well-defined paths with minimal loss. This resilience significantly enhances sensitivity and ensures stable device performance, even in the presence of fabrication imperfections or challenging conditions. These characteristics make topological photonic platforms particularly well-suited for applications requiring high precision, reliability, and environmental robustness.

Among the various topological photonic platforms, valley photonic crystals (VPCs) have emerged as a promising platform for topological photonic integration,^[15–19] offering robust sensing solutions. VPC-based sensing chips, particularly those featuring waveguide-cavity integration, exhibit sharp and stable transmission dips with reduced scattering losses and a larger interaction region, all of which collectively enhance detection sensitivity.^[20–22] Furthermore, the on-chip VPC system avoids the high path loss typically observed in conventional free-space THz measurements. VPC devices are often fabricated on

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silicon (Si), making them inherently compatible with standard complementary metal-oxide-semiconductor (CMOS) processing technologies,^[15] which is essential for scalable manufacturing and integration with existing electronic systems. Despite their advantages, conventional VPC sensors face significant challenges in realizing compact, high-sensitivity on-chip systems. Foremost among these is the inherently short light-matter interaction time, which limits their sensing performance. To compensate, larger cavity dimensions are often required to increase the optical path length, thereby enhancing the light-matter interaction time. However, this approach directly conflicts with the miniaturization goals of VPC sensors. A promising solution lies in the use of slow light within these topological photonic platforms.^[23–28] Slow-light, characterized by a significantly reduced group velocity, enhances light-matter interaction time without increasing the device footprint. As such, slow-light VPCs can simultaneously offer higher sensitivity and reduction in round-trip path lengths and cavity dimensions compared to their non-slow-light counterparts, two key requirements for next-generation on-chip sensing technologies. Thus, it becomes highly relevant to investigate the impact of slow-light phenomena on the sensing performance of topological waveguide-cavity systems operating in the THz range.

In this work, we demonstrate a significant enhancement in the sensitivity of VPC waveguide-cavity systems enabled by slow-light effects. Specifically, we introduce a slow-light regime at the Brillouin zone edge of a bearded-type interface rather than the commonly used zigzag-type interface in topological cavity designs. Building on this concept, we design three distinct cavity configurations: a non-slow-light cavity and two slow-light cavities with varying zigzag-to-bearded interface section lengths and different round-trip path lengths. We further analyze the spatial extent of the evanescent field and the evanescent field ratio (EFR) to show that a significant portion of the optical energy resides in the evanescent tail of the cavity modes, crucial for evanescent field-based sensing. Additionally, comparable values of the spatial extent and the EFR indicate that a physical reduction in the cavity footprint can be effectively compensated through dispersion and modal control. Our results show that, despite their shorter optical path lengths, the slow-light cavities possess external evanescent field confinement factors comparable to their non-slow-light counterpart, indicating enhanced light-matter interaction time. The polyimide (Kapton) thin-film thickness sensing experiments are performed to evaluate the sensing performance of the three VPC cavities. Kapton is chosen for its refractive index (~ 1.6),^[8] which is comparable to that of biomolecules such as DNA and RNA (typically ranging from 1.4 to 1.7).^[29,30] By measuring the resonant frequency shifts, we demonstrate that slow-light VPC cavities exhibit up to a threefold increase in detection sensitivity compared to the non-slow-light cavity. This sensitivity enhancement corresponds to a threefold improvement in the thickness resolution. These results highlight the potential of slow-light VPCs as a platform for next-generation, high-performance THz sensing. Our findings pave the way for more reliable and accurate THz sensing technologies, pushing the boundaries of photonics-based approaches for environmental monitoring, biosensing, and chemical analysis.

2. Results and Discussion

The topological sensing chips are fabricated on a high-resistivity ($>10 \text{ k}\Omega \text{ cm}$) Si (HR-Si) wafer, chosen for its CMOS compatibility, low absorption loss, and high refractive index at THz frequencies (see Experimental Section for fabrication details).^[20,21] Each chip integrates topological waveguides coupled to topological cavities, as shown in the optical image of a representative THz topological waveguide-cavity chip in **Figure 1a**. The VPC unit cells in these waveguide-cavity chips comprise a honeycomb lattice (lattice constant $a = 242.5 \text{ }\mu\text{m}$), each featuring two inverted equilateral triangular air holes with side lengths l_1 and l_2 . Unit cells are categorized as type A ($l_1 = 1.3a/\sqrt{3} > l_2 = 0.7a/\sqrt{3}$) and type B ($l_1 = 0.7a/\sqrt{3} < l_2 = 1.3a/\sqrt{3}$), as illustrated in the inset of **Figure 1a** (type A and B outlined in green and yellow dashed lines, respectively). These unit cells are used to construct zigzag- and bearded-type BA interfaces (indicated by white and blue dashed lines, respectively). The chip thickness is $220 \text{ }\mu\text{m}$ to ensure strong vertical confinement of THz waves via total internal reflection. A taper at the input end couples the THz signal into the chip through a topological valley-locked waveguide, enabling robust transmission through sharp corners with bending radii of $r = 0 \text{ }\mu\text{m}$.^[31] The inset in **Figure 1a** shows the magnified optical image of the taper of length 3.0 mm and width, $w = 242.5 \text{ }\mu\text{m}$. Additionally, the bending loss per turn is numerically calculated to be $\sim 0.1 \text{ dB turn}^{-1}$ for one of the designed slow-light devices (see **Section S7**, Supporting Information). Thus, although the actual bending loss values of the fabricated devices may be higher than the numerically calculated value, the comparatively low bending loss is still confirmed, demonstrating the robustness of the slow-light mode. The topological cavity, coupled to the waveguide, traps a portion of the THz energy, while the rest is coupled out of the chip through the output taper, with identical dimensions to the input taper. When an analyte, such as a polyimide film, is placed on the chip, the cavity's evanescent field interacts with it, resulting in a measurable shift in the resonance frequency, which serves as an indicator of analyte detection.

When the VPC unit cells have equal triangular air holes ($l_1 = l_2$), the in-plane inversion symmetry is preserved due to the inherent C_6 symmetry of the lattice, resulting in degenerate Dirac points at the K and K' valleys (red dashed line in **Figure 1b** (i)). Breaking this symmetry by setting $l_1 \neq l_2$ reduces the lattice symmetry to C_3 , lifting the Dirac point degeneracy and opening a topological bandgap (blue solid line in **Figure 1b** (i)). This induces opposite rotational power flows in type A and type B unit cells, visualized by the contrasting Poynting vector flow vortices (black arrows in **Figure 1b** (ii, iii)). Specifically, at the K valley, the type A unit cell exhibits clockwise Poynting vector flow vortices, and the type B unit cell shows counterclockwise Poynting vector flow vortices, resulting in distinct topological phases of the out-of-plane magnetic field (ϕ_H), each completing a full 2π rotation around the smaller triangular airhole. These phases correspond to opposite valley Chern numbers for type A and type B unit cells.^[15–17] By interfacing type A and B VPCs, topological edge states are formed due to bulk-boundary correspondence.^[32] For the topological sensing chips, two types of interfaces are considered: zigzag- and bearded-type BA interfaces, both designed

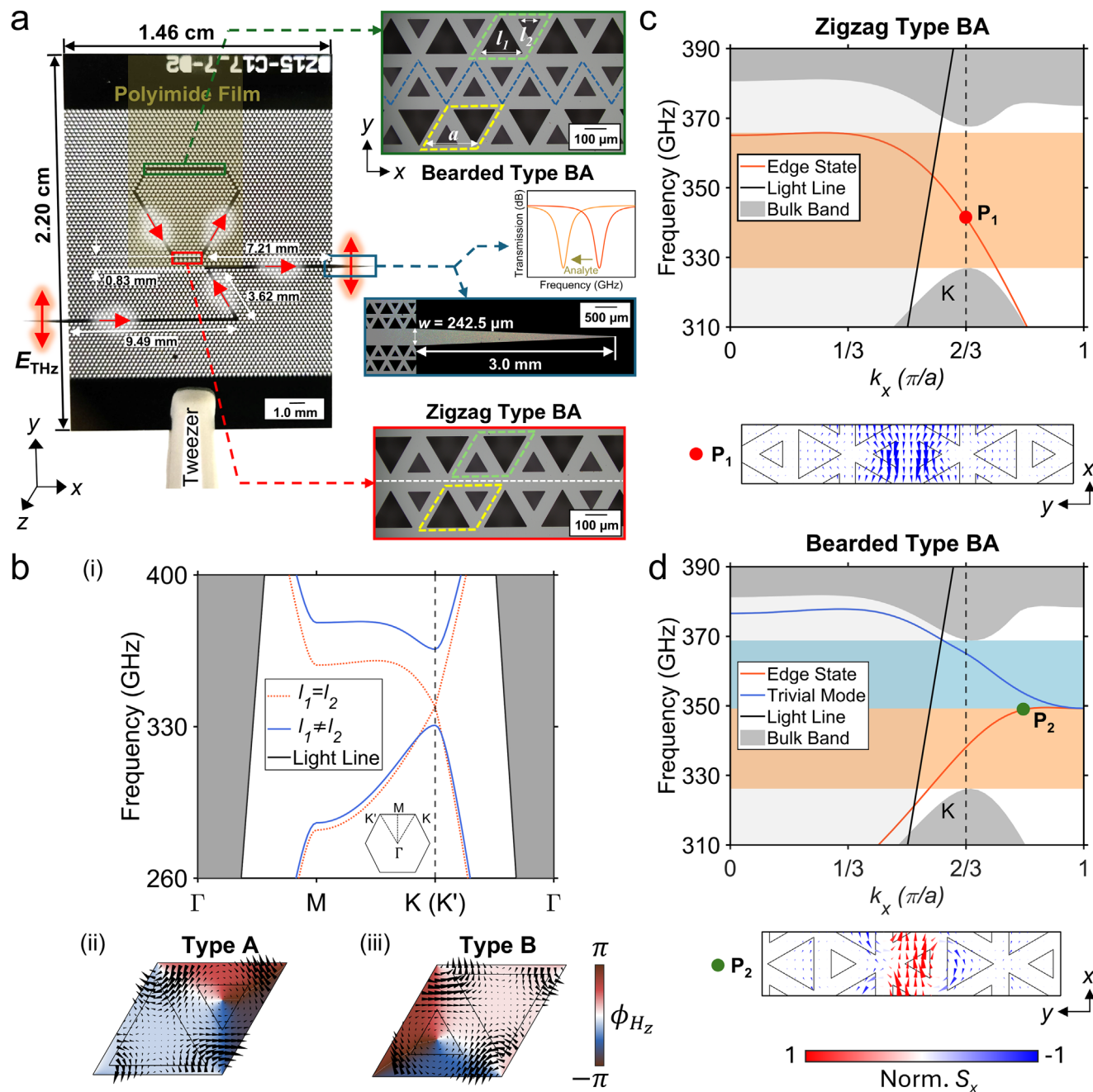


Figure 1. On-chip sensing using a THz slow-light topological chip. a) Optical image of the THz topological waveguide-cavity chip with a 1.46 cm × 2.20 cm footprint. The device features a bearded-type BA topological waveguide coupled to a topological cavity composed of both zigzag- and bearded-type BA interfaces (highlighted in red and green boxes, respectively). The inset shows a magnified optical image of the zigzag- and bearded-type BA domain walls (in white and blue, respectively), along with the type A (green) and type B (yellow) VPC unit cells. Each unit cell consists of two equilateral triangular air holes with l_1 and l_2 side lengths. The lattice constant is $a = 242.5 \mu\text{m}$; for the type A unit cell, $l_1 = 1.3a/\sqrt{3}$ and $l_2 = 0.7a/\sqrt{3}$, whereas for the type B unit cell, $l_1 = 0.7a/\sqrt{3}$ and $l_2 = 1.3a/\sqrt{3}$. The analyte (polyimide film) is placed on the chip to cover the entire cavity region. The THz signal is coupled into and out of the chip using two tapers at opposite ends. The inset shows the magnified optical image of the taper of length 3.0 mm and width, $w = 242.5 \mu\text{m}$. The transmission spectrum measured at the output taper exhibits a shift in the cavity resonance upon introducing the analyte. b) Simulated band structure of the VPC unit cell with (i) preserved ($l_1 = l_2$) and broken ($l_1 \neq l_2$) inversion symmetry. The phase distributions of the magnetic field (ϕ_{H_z}) for the lower bulk band of the (ii) type A and (iii) type B VPC unit cells at the K valley are shown. Black arrows indicate the Poynting vector flow, which is clockwise in (ii) and counterclockwise in (iii) at the K valley. Simulated projected band diagrams of the c) zigzag-type BA and d) bearded-type BA interfaces. The bottom panels in (c) and (d) show the Poynting vector flow corresponding to the edge states at the two marked points: P_1 (red dot) and P_2 (green dot) in the respective band diagrams. The red and blue arrows indicate the Poynting vectors of forward-propagating (+x direction) and backward-propagating (−x direction) waves, respectively, associated with the topological edge states.

with smaller airholes at the boundary. The zigzag-type BA interface supports non-slow-light modes, evidenced by the nearly linear dispersion of the topological edge state (red solid line in Figure 1c) near the K valley, extending from 327–355 GHz within the lower-frequency portion of the topological bandgap (orange-shaded region). The edge state becomes less dispersive but highly radiative beyond 355 GHz, above the light line (black solid line). In contrast, reducing the air hole size near the bearded-type BA interface increases the local refractive index, shifting the topological edge state (red line in Figure 1d) to lower frequencies and giving rise to a trivial mode (blue solid line) at the upper band edge. Due to glide-plane symmetry, both the topological and trivial modes become degenerate at the Brillouin zone (BZ) edge.^[23–25] This results in near non-dispersive edge states near the topological-trivial bandgap boundary, spanning 326–349 GHz and 349–369 GHz (orange and blue shaded regions in Figure 1d), characteristic of slow-light behavior. Crucially, unlike the zigzag-type BA interface, the bearded-type BA interface remains non-radiative, residing below the light line. The bottom panels of Figure 1c, d illustrate the Poynting vector flows at the points P_1 and P_2 in the projected band diagrams of the zigzag and bearded interfaces, respectively (see Section S5, Supporting Information, for a more in-depth study of the Poynting vector flows). The red and blue arrows represent the Poynting vectors of forward (+x direction) and backward (−x direction) propagating waves associated with the topological edge states. The Poynting vector flow at P_1 (red dot, zigzag interface) indicates a pure backward-propagating topological edge state at the K valley. In contrast, at P_2 (green dot, bearded interface), the Poynting vector flow reveals a forward-propagating wave along the interface, accompanied by two backward-propagating waves in the adjacent VPC domains.

The non-slow-light and slow-light modes, originating from the zigzag- and bearded-type BA interfaces, respectively, play a crucial role in shaping the three-dimensional (3D) field distribution, a key parameter for sensing. To investigate this, three distinct cavity configurations are designed and analyzed. The first configuration, a non-slow-light (NSL) cavity (Figure 2a), is constructed only using zigzag-type BA interface, with a length of $L = 30a$ and an optical path length of $90a$. The other two configurations are slow-light cavities that incorporate both zigzag- and bearded-type BA interface sections. In the first slow-light cavity (SL_1) (Figure 2b), the zigzag and bearded sections have lengths $L_1 = 11a$ and $L_2 = 6a$, respectively. The second slow-light cavity (SL_2) (Figure 2c) features a longer round-trip path and a greater contribution from the bearded interface, with $L_1 = 6a$ and $L_2 = 18a$. The optical path lengths are $51a$ and $72a$ for the SL_1 and SL_2 cavities, respectively (see Section S6, Supporting Information, for design details on the efficient guidance of THz waves through the hybrid interfaces). 3D simulations are performed (see Experimental Section for numerical calculation details) to compute the normalized electric field distributions (Norm. $|E|$) of the cavity eigenmodes in both the in-plane (x - y) and out-of-plane (x - z) directions, as shown in Figure 2a–f, respectively. Across all configurations, including those incorporating both zigzag- and bearded-type BA interfaces, the simulated topological modes are observed to propagate robustly through sharp bends, thereby facilitating the formation of whispering gallery-like modes within the cavities.^[24]

In the NSL cavity, the resonance mode is formed due to the non-slow-light topological edge state. The in-plane Norm. $|E|$ distribution at the resonance frequency of 350 GHz (Figure 2a) shows strong confinement of the mode along the topological interface throughout the entire cavity round-trip path. In contrast, the SL_1 and SL_2 cavities support resonance modes formed through the hybridization of the non-slow-light zigzag interface mode with the slow-light mode of the bearded interface, which has a lower group velocity (v_g). The in-plane Norm. $|E|$ distributions in SL_1 and SL_2 cavities at their respective resonance frequencies of 345 and 339 GHz (Figure 2b,c) exhibit strong confinement and robustness to sharp turns. The spatial extent of these cavity modes is crucial for sensing applications, as the exponentially decaying evanescent field in the z -direction interacts with analytes. The out-of-plane (x - z) Norm. $|E|$ distributions along the cross-sections highlighted in Figures 2a–c are shown in Figures 2d–f, revealing a considerable amount of mode energy residing in the evanescent field tails. This enhances the sensing capability of cavity modes. The out-of-plane interaction is quantified using the evanescent field ratio (EFR), defined as the ratio of the energy in the evanescent tail to the total energy^[20,33] (see Section S1, Supporting Information).

$$EFR = \frac{\int_{\text{evanescent}} |E|^2 dV}{\int_{\text{total}} |E|^2 dV} \quad (1)$$

here, E represents the electric field distribution. The EFRs for the NSL, SL_1 , and SL_2 cavities are 8.19%, 8.25%, and 8.17%, respectively. Additionally, the spatial extent of the evanescent field (z_e) is defined as the distance from the surface of the VPC chip at which the electric field amplitude decays to $1/e$ ($\approx 36.7\%$) of its initial value. Figures 2g–i show exponential fits to the out-of-plane Norm. $|E|$ profiles calculated along the cross-sections highlighted in Figures 2a–c. The insets in each panel indicate the corresponding z_e values for the different sensing chips. Notably, all of the topological cavity modes, whether non-slow-light or slow-light, exhibit nearly identical EFRs and evanescent field extents. The similarity of these values between the smaller slow-light cavities and the larger non-slow-light cavity indicates that slow-light engineering effectively compensates for reduced physical dimensions through dispersion and modal control.

To quantify the slow-light-enhanced light-matter interaction via the evanescent field, the external evanescent field confinement factor (Γ) is calculated, which is defined as:^[34,35]

$$\Gamma = n_g/n_{cl} \cdot f = \frac{n_g}{n_{cl}} \frac{\int_{cl} \epsilon |E|^2 dV}{\int_v \epsilon |E|^2 dV} \quad (2)$$

here, n_g is the group index, n_{cl} is the refractive index of the cladding, and f is the fraction of field energy overlapping with the cladding (see Section S1, Supporting Information). To maximize detection sensitivity, both f and n_g should be as large as possible. The resulting values of Γ are calculated to be 8.56% for the NSL cavity, and 9.76% and 8.65% for the SL_1 and SL_2 cavities, respectively. These results highlight that, despite significantly shorter optical path lengths, the slow-light cavities exhibit even stronger light-matter interactions than the non-slow-light cavity. To validate this enhanced interaction in the

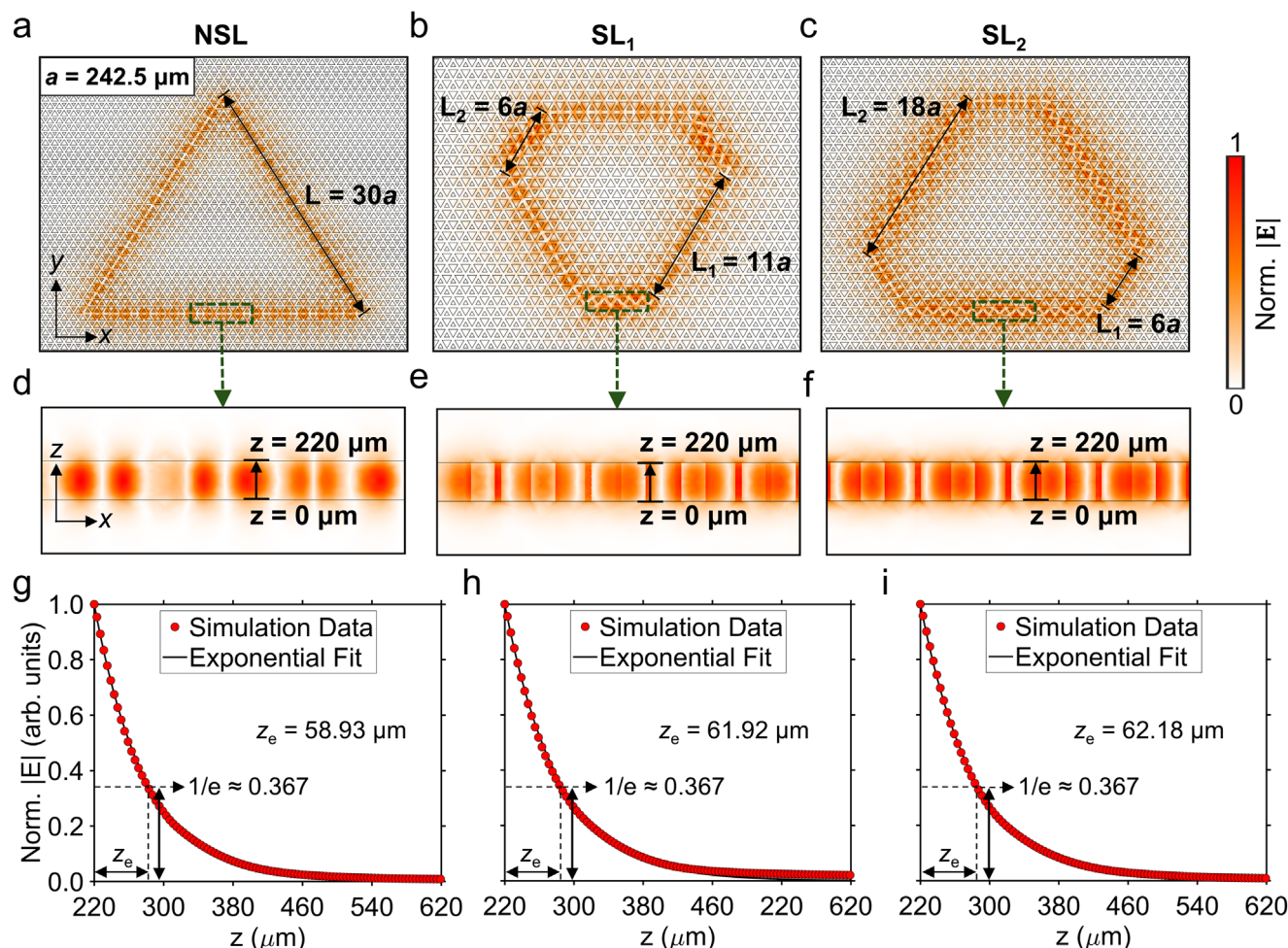


Figure 2. E-field distributions and spatial extent of evanescent fields of non-slow-light and slow-light topological VPC cavities. Simulated in-plane (x - y plane) normalized electric field, Norm. |E|, distributions for three topological VPC cavity configurations: a) a non-slow-light (NSL) cavity consisting of a zigzag-type BA interface with length $L = 30a$, b) a slow-light cavity (SL₁) composed of zigzag- and bearded-type BA interfaces with lengths $L_1 = 11a$ and $L_2 = 6a$, respectively, and c) a slow-light cavity (SL₂) with $L_1 = 6a$ and $L_2 = 18a$. The resonance frequencies for the cavities in (a), (b), and (c) are 350 GHz, 345 GHz, and 339 GHz, respectively. Simulated out-of-plane (x - z) views of the Norm. |E| fields, taken along the highlighted (dark green) cross-sections in (a)–(c), are shown in (d)–(f), respectively. Also indicated in (d)–(f) are the bottom surface of the cavities at $z = 0 \mu\text{m}$, and the top surface at $z = 220 \mu\text{m}$, for both the NSL and SL cavity configurations. Exponential fits to the Norm. |E| field data in the highlighted cross-sections from the top surface at $z = 220 \mu\text{m}$ of (a)–(c) for the g) NSL, h) SL₁, and i) SL₂ VPC cavities. The spatial extents of the evanescent field (z_e) are indicated in each panel corresponding to the point along the z -axis at which Norm. |E| = 1/e.

slow-light cavities, waveguide-cavity VPC chips are designed based on the NSL, SL₁, and SL₂ cavity configurations (see Section S6, Supporting Information, for more details on the design of NSL and SL waveguide-cavity integrated chips), and Kapton thin-film sensing is performed to compare the performance of the non-slow-light and slow-light cavities. The non-slow-light VPC chip is implemented with a zigzag-type BA topological waveguide to excite the NSL cavity, whereas the slow-light VPC chips utilize bearded-type BA topological waveguides to excite the corresponding hybrid cavities (SL₁ and SL₂). Kapton thin films of varying thicknesses (8–25 μm) are placed over the cavity regions. As the thickness increases, a clear redshift in the cavity resonance frequency is observed (Figures 3a–c), caused by the perturbation of the local refractive index due to the analyte's interaction with the evanescent field (see Section S1, Supporting Information). By fitting the transmission spectra of the cavity resonances without the

Kapton thin-film on top using the Fano equation, the linewidths of the cavity resonances (δf) are calculated to be 122.95, 242.30, and 250.19 MHz at resonance frequencies of 349.41, 347.24, and 340.50 GHz for NSL, SL₁, and SL₂ cavities, respectively (see Section S3, Supporting Information). The cavity modes are chosen based on their nearly identical transmission dips, indicating comparable confined energies within the cavities. This allows for a fair comparison of the sensing performance across different cavity types. Figure 3d presents the optical images of the fabricated non-slow-light and slow-light sensing chips. The metallic hollow waveguide units, labeled W_1 and W_2 , are employed to couple THz waves into and out of the chips. Figure 3e shows the resonance frequency shift (Δf) as a function of increasing Kapton thin-film thickness (t). Since the film thickness is comparable to the spatial extent of the localized electric field (z_e), an exponential fit of Δf is performed (see Section S4, Supporting Information). The

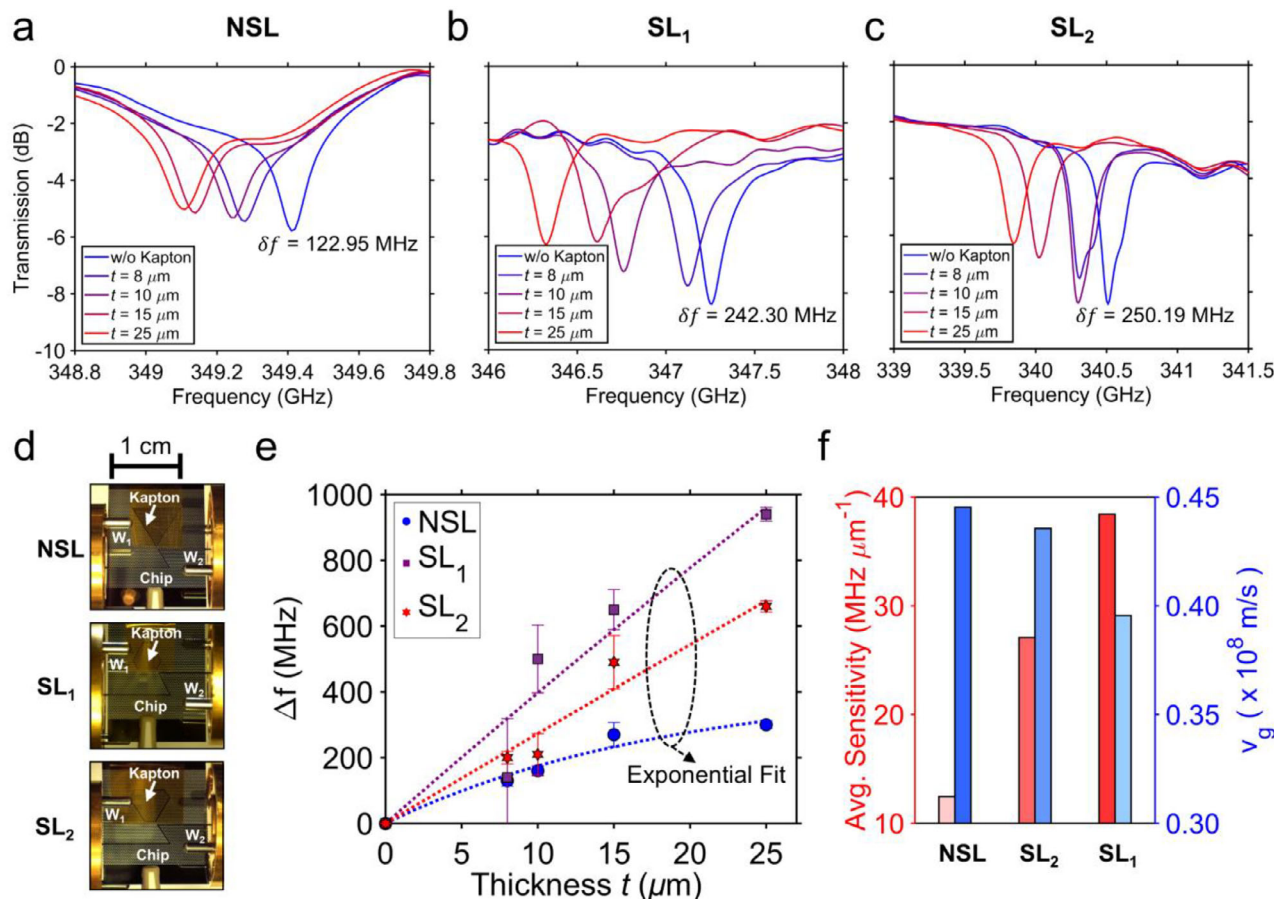


Figure 3. Enhanced thickness sensing of Kapton thin-films using slow-light topological waveguide-cavity VPC chips. Measured transmission spectra of three VPC chips incorporating a) NSL, b) SL_1 , and c) SL_2 cavities, recorded for different Kapton thin-film thicknesses (t). The resonance linewidths (δf) are indicated in each panel. d) Optical images of the fabricated VPC chips and the experimental setup. The Kapton thin film is placed directly on top of the cavity region of each chip. W_1 and W_2 represent the metallic hollow waveguide units that couple the THz signal into and out of the chip. e) Measured shifts in cavity resonance frequency (Δf) as a function of Kapton thin-film thickness (t) for the three VPC configurations. Dashed lines denote exponential fits to the experimental data. f) Calculated average sensitivities and corresponding group velocities (v_g) of the excited cavity modes for the VPC chips shown in (d).

results show that for identical Kapton thin-film thicknesses, the slow-light cavities (SL_1 and SL_2) exhibit larger Δf shifts than the non-slow-light NSL cavity. To quantify this, the sensitivity is defined as $S = \Delta f / \Delta t$, where Δf is the frequency shift and Δt is the change in analyte thickness. We calculate the average sensitivity (S_{avg}) values for the VPC chips (see Section S4, Supporting Information). Figure 3f compares the calculated average sensitivity (red bars) and group velocity (v_g , blue bars) of the cavity modes. The slow-light cavities exhibit up to threefold higher sensitivity compared to the NSL cavity, which has an S_{avg} of $12.46 \text{ MHz } \mu\text{m}^{-1}$. Furthermore, this trend of increasing sensitivity corresponds to the lowering of group velocities due to slow-light effects. Among the slow-light designs, SL_1 shows a higher S_{avg} of $38.42 \text{ MHz } \mu\text{m}^{-1}$ compared to $27.09 \text{ MHz } \mu\text{m}^{-1}$ for the SL_2 cavity, attributed to the relatively slower v_g of the excited mode in the smaller cavity, which strengthens its interaction with the analyte. The thickness resolution $D_t = D_f / S_{\text{avg}}$ defines the minimum change in thickness that can be distinguished, where D_f is the resolution in the frequency domain of the measurement setup. In our experimental setup, the chosen D_f is 10 MHz, from

which D_t for the NSL and SL VPC chips is calculated. Specifically, the NSL chip has a D_t of $0.80 \text{ } \mu\text{m}$, which is higher than the D_t of 0.26 and $0.37 \text{ } \mu\text{m}$ for the SL_1 and SL_2 VPC chips, respectively. This shows that the slow-light cavities can detect thickness changes up to three times smaller than those detectable by the NSL cavities using an identical experimental setup. Finally, we calculate the detection factor (P) figure-of-merit (FOM) for all our VPC chips, defined as $P = \text{FWHM} / S$, where FWHM is the linewidth of the cavity resonance, and S is the sensitivity (taken as S_{avg} in our calculations). For the NSL cavity, we calculate $P = 9.86 \text{ } \mu\text{m}$, which is higher or comparable to the values of 6.31 and $9.24 \text{ } \mu\text{m}$ obtained for the SL_1 and SL_2 cavities, respectively. Generally, sensor designs with lower detection factors are more desirable.^[36] Thus, these findings highlight the feasibility of using topological slow-light cavities for enhanced sensing performance in VPC-based platforms.

In summary, we experimentally demonstrated and compared two slow-light VPC sensing chips with a non-slow-light counterpart, all fabricated on a silicon photonics platform. These sensing chips were designed by leveraging topological edge states

supported at the zigzag- and bearded-type BA interfaces. Owing to the topological robustness of these edge states, all the topological cavities exhibit robust transmission even in the presence of sharp bends. We showed that slow-light engineering of the cavity designs effectively compensates for the reduction in cavity footprint through dispersion and modal control, resulting in stronger light-matter interactions within smaller cavity dimensions. We performed Kapton thin-film thickness sensing experiments and demonstrated that the slow-light VPCs achieve a threefold enhancement in sensitivity and a threefold reduction in thickness resolution compared to the non-slow-light chip. Our results show that slow-light VPC waveguide-cavity systems can effectively address key limitations in on-chip THz sensing, notably enhancing sensitivity while minimizing cavity footprint. By leveraging topological robustness to enhance reliability and accuracy, these systems broaden the potential of photonics-based technologies, with potential applications in miniaturized lab-on-chip sensing.

3. Experimental Section

Sample Fabrication: The VPC chips were fabricated on a 4-inch high-resistivity ($>10\text{ k}\Omega\text{ cm}$) silicon (HR-Si) wafer. The wafer was first thoroughly cleaned, followed by the deposition of a $2\text{ }\mu\text{m}$ thick silicon dioxide (SiO_2) layer. Photolithography was used to pattern the triangular etch holes and device separation trenches. The SiO_2 layer was then selectively etched to expose the underlying silicon surface using reactive ion etching (RIE). Finally, deep reactive ion etching (DRIE) was carried out to a depth of $220\text{ }\mu\text{m}$.

Experimental Setup: The Vector Network Analyzer (VNA) measurement setup utilized a Keysight N5222B network analyzer, which generated microwave signals in the 10–26.5 MHz range. These signals were up-converted to the WR-2.2 frequency range (320–500 GHz) using a Virginia Diodes Inc. (VDI) WM-570 VNAX frequency extension module. The resulting THz signal was coupled into the VPC chip via a WR-2.2 metallic hollow waveguide through the input taper. The transmitted THz signal was collected using another WR-2.2 waveguide and directed into a second VDI WM-570 VNAX module, where it was down-converted to an intermediate frequency (IF) using a local oscillator (LO) signal generated by the network analyzer. The IF signal was then analyzed to characterize the transmission response of the VPC chip. Prior to measurement, the system was calibrated using the SOLT (Short-Open-Load-Through) waveguide calibration procedure, following WR-2.2 standards. This calibration compensated for system non-idealities and ensured accurate measurement of the VPC chip's response.

Numerical Calculation of the Band Structure and Cavity Eigenmodes: Two-dimensional (2D) simulations of the VPC unit cell band structure and edge states were conducted using the Eigenmode solver in COMSOL Multiphysics. To approximate three-dimensional (3D) behavior in 2D, an effective index of 3.0 was used for silicon. For the band diagram of the VPC unit cell and the edge states of the topological interface, Floquet boundary conditions were applied along both the x - and y -directions for the unit cell, and along the x -direction for the topological interface simulations. Additionally, scattering boundary conditions were applied along the y -direction in the topological interface simulations.

3D simulations of the topological cavity eigenmodes were performed using CST Microwave Studio. The dielectric constants were set to 1.0 for air and 11.7 for silicon. The 3D cavity structure includes 16 unit cells in the outer VPC regions from the boundary between the two distinct VPC bulks. Open (add space) boundary conditions were applied to eliminate unwanted reflections and artifacts. Additionally, a plane of symmetry was imposed along the central plane of the cavity to reduce computational cost. The Eigenmode solver in CST was used for simulation, and the 3D electric field (E-field) distributions of the selected cavity modes were extracted for analysis.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

B.C. and R.S. conceived the idea; B.C. performed the numerical analysis with the help of W.W. and N.N.; B.C. performed the on-chip transmission measurement with the help of W.W., N.N., and Y.J.T.; B.C. wrote the initial draft of the manuscript; W.W., N.N., Y.J.T., and R.S. revised the manuscript; R.S. led the overall project.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

on-chip sensor, slow light, THz sensing, topological photonics, topological slow light cavities, topological waveguides

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